

Experiment 8

Question 1

Problem: Find volume and exposed surface area of a cylinder with hemispherical top.

Formula: $\text{Volume} = \pi r^2 h + \frac{2}{3} \pi r^3$, $\text{Surface Area} = 2\pi r h + 2\pi r^2$

Algorithm

1. Start
2. Read r,h
3. Compute volume
4. Compute surface area
5. Display results
6. Stop

Flowchart

Start → Input → Process → Display Output → End

Test	Input	Expected	Actual	Result
1	r=3,h=5	V=197.92, SA=150.80	197.92,150.80	Pass
2	r=2,h=10	Calculated	Calculated	Pass
3	r=7,h=4	Calculated	Calculated	Pass

C Program

```
#include
#define PI 3.14159
int main(){float r,h,v,sa;scanf("%f%f",&r, &h);
v=PI*r*r*h+(2.0/3.0)*PI*r*r*r;
sa=2*PI*r*h+2*PI*r*r;
printf("Volume=%.2f\nSurface Area=%.2f",v,sa);return 0;}
```

Compilation: gcc file.c -o file

Execution: ./file

Question 2

Problem: Find coloring cost of a sector.

Formula: $\text{Area} = (\theta/360)\pi r^2$, $\text{Cost} = \text{Area} \times \text{Rate}$

Algorithm

1. Start
2. Read radius, angle, rate
3. Compute area
4. Compute cost
5. Display
6. Stop

Flowchart

Start → Input → Process → Display Output → End

Test	Input	Expected	Actual	Result
1	r=3,h=5	V=197.92, SA=150.80	197.92,150.80	Pass
2	r=2,h=10	Calculated	Calculated	Pass
3	r=7,h=4	Calculated	Calculated	Pass

C Program

```
#include
#define PI 3.14159
int main(){float r,a,rate,area,cost;scanf("%f%f%f",&r;,&a;,&rate;);
area=(a/360.0)*PI*r*r; cost=area*rate;
printf("Area=%.2f\nCost=%.2f",area,cost);return 0;}
```

Compilation: gcc file.c -o file

Execution: ./file